Five summands in the iid Berry–Esseen conjecture

A sign-count proof and the finite-index dependency chain

Kacper Rodziewicz Bartosz Kołodziejek

Working draft — 20 October 2026

A revised and corrected version is intended for submission to arXiv.

Abstract

Using Prawitz's smoothing inequality and Shevtsova's iid estimates, we prove $C_5 \le C_{\mathrm{E}} = (3+\sqrt{10})/(6\sqrt{2\pi})$ from the published four-summand bound. A fixed-index extremal argument gives an attained closed-CDF witness and bounded support before higher moments are used. Complete moment and sign-probability covers reduce the remaining gapped regions to exact rational count inequalities. A direct conditional-binomial and Cantelli comparison treats the final corner without a support-gap, Fourier or predecessor assumption. Endpoint atoms, closed seams and both infinite threshold rays are retained throughout. The fifth-index argument uses no eventual-in-n theorem. Under separately stated analytic and historical continuation hypotheses, a finite chain also gives $C_n < C_{\mathrm{E}}$ for every $n \ge 6$; these hypotheses are kept explicit in the dependency diagram. ChatGPT and Codex assisted the development of the proofs and computational certificates and the preparation of this manuscript, using the GPT-5.6 Sol, GPT-6.1 Sol and GPT-6 Astra models.

1 Statement and precise scope

For a real law P with $\mathbb{E}X = 0$, $\mathbb{E}X^2 = 1$ and $\rho = \mathbb{E}|X|^3 < \infty$, let

$$R_n(P) = \frac{\sqrt{n}}{\rho} \sup_{t\in\mathbb{R}} \left|\mathbb{P}(X_1 + \cdots + X_n \le t) - \Phi(t/\sqrt{n})\right|, \qquad C_n = \sup_P R_n(P),$$

where the X_i are independent with the *same* law P. Write

$$C_{\mathrm{E}} = \frac{3+\sqrt{10}}{6\sqrt{2\pi}} = 0.40973218\ldots.$$

We prove the unrestricted assertion $C_5 \le C_{\mathrm{E}}$ from the four-summand input. We use the published bound $C_4 \le C_{\mathrm{E}}$ from [1, Theorem 26.1, p. 174]. That theorem states the stronger $C_4 \le 0.4097$; the stronger value is not needed here.

Theorem 1. *Assume $C_4 \le C_{\mathrm{E}}$. Then $C_5 \le C_{\mathrm{E}}$.*

This is a computer-assisted proof. Its continuous inequalities, finite certificates, and complete domain coverage are separate proof obligations. The final verification record identifies the exact successful proof objects, their domains, and their input files. The continuous support, Fourier, resource and count arguments are supplied in the companion proof sources indexed in Section 8. This note explains the composition and proves the new gap-free part in detail; it does not replace those companion proofs by numerical success messages. No eventual-in-n theorem or monotonicity of C_n is used.

Besides $C_4 \le C_{\mathrm{E}}$, the cited inputs are Prawitz smoothing in the normalization of [2, Lemma 4.1, p. 262] and the iid bounds of [3, p. 125]. The continuous reductions and exact arithmetic certificates are supplied in the accompanying proof supplement [4].

2 Why a contrary law is a legitimate object

Lemma 2 (Fixed-index attainment and the correct jump edge)**.** *For every fixed integer $n \geq 1$, C_n is attained. After reflection of the whole marginal if necessary, there are an attaining standardized law P and a finite raw threshold H such that*

$$C_n = \frac{\sqrt{n}}{\rho}\{\mathbb{P}(S_n \leq H) - \Phi(H/\sqrt{n})\} > 0.$$

Proof. The symmetric two-point law gives a positive lower bound for C_n, while $R_n(P) \leq \sqrt{n}/\rho$. Consequently a maximizing sequence has bounded third absolute moments. It is tight. On a weakly convergent subsequence, first and second moments pass to the limit, since their tail expectations are at most M/L^2 and M/L, respectively. If $\rho_j \to r$, lower semicontinuity gives $\rho(P) \leq r$. No uniform integrability of third moments has been assumed.

Because n is fixed, the n-fold product laws, and hence their sums, converge weakly. All normalized sums have variance one, so near-maximizing thresholds stay bounded by Chebyshev's inequality and the Gaussian tails. For a convergent sequence of thresholds $z_j \to z$ the exact bounds are

$$\limsup F_j(z_j) \leq F(z), \qquad \liminf F_j(z_j) \geq F(z-).$$

Both right and left jump errors are bounded by the Kolmogorov distance from the continuous Gaussian CDF. Thus $\limsup \Delta_n(P_j) \leq \Delta_n(P)$. It follows that

$$C_n r \leq \sqrt{n}\Delta_n(P) \leq C_n \rho(P) \leq C_n r.$$

Equality proves attainment, including equality of the limiting third moments.

The functions $F(z) - \Phi(z)$ and $\Phi(z) - F(z-)$ are upper semicontinuous and vanish at infinity. Their positive maxima are attained. Reflection converts the second branch exactly into the first, because $F_{-S}(-H) = 1 - F_S(H-)$. The endpoint atom is retained. □

Assume now that $C_5 > C_{\mathrm{E}}$, and choose the witness from Lemma 2. For this same marginal its predecessor score satisfies $R_4(P) \leq C_4$. The contact allowance is therefore

$$q_5 = 5\big(R_4(P) - C_5\big)_+ = 0. \tag{1}$$

Equation (1) is the only use of the four-summand input. It does not assert $C_5 \leq C_4$.

2.1 Contact before any higher moment is used

Put $z = H/\sqrt{5}$, $F = \mathbb{P}(S_5 \leq H)$, $a = (F - \Phi(z))/\rho > 0$, $\beta = \mathbb{E}[X|X|]$, and $h(x) = \mathbb{P}(S_4 \leq H - x)$. Contaminate the common marginal by an atom at an arbitrary real x and then center and scale the entire mixture. The raw event $S_5 \leq H$ is kept fixed. Its first derivative is $5(h(x) - F)$, so atoms cause no CDF differentiation issue. The central third moment derivative is $|x|^3 - \rho - 3x\beta$, with the variance normalization adding $-3\rho(x^2 - 1)/2$. Global maximality yields

$$5h(x) \leq a|x|^3 + q_0 + q_1 x + q_2 x^2, \tag{2}$$

where

$$q_0 = 5F + \tfrac{1}{2}(a\rho + z\phi(z)), \quad q_1 = -\sqrt{5}\phi(z) - 3a\beta, \quad q_2 = -\tfrac{1}{2}(3a\rho + z\phi(z)).$$

The nonnegative slack has expectation zero. It is lower semicontinuous, so equality holds at every support point. Since the cubic coefficient a is positive and $0 \leq 5h \leq 5$, the support is bounded. Higher moments are used only after this conclusion.

Quantitative support follows by inserting the all-threshold predecessor bound into this contact equality and using (1). Its numerical certificate covers both support rays and the whole threshold line, including both infinite threshold tails. A finite mesh without its curvature remainder would not be sufficient.

3 The complete moment-domain cover

Let $\beta = \mathbb{E}[X|X|]$. An elementary moment restriction is

$$\rho \geq \frac{1+\beta^2}{\sqrt{1-\beta^2}}. \tag{3}$$

Indeed, if $m = \mathbb{E}X_+ = \mathbb{E}X_-$ and $a_\pm = (1 \pm \beta)/2$, signwise Cauchy inequalities give $m \leq \sqrt{a_+ a_-}$ and $\rho \geq (a_+^2 + a_-^2)/m$. Define

$$b(r) = \left(\frac{r\sqrt{r^2+8} - r^2 - 2}{2} \right)^{1/2}.$$

Every law with $\rho \leq r$ satisfies $|\beta| \leq b(r)$. The certificates use the rational upper enclosures

$$b_2 = \frac{84729828287999}{250000000000000} \geq b(237/200), \qquad b_3 = \frac{35139006743001}{10^{14}} \geq b(6/5).$$

These enclosing inequalities are checked exactly.

The low-moment cover is as follows. Each gapped row has a complete 100-panel partition of $p = \mathbb{P}(X > 0)$; the last column is the number of panels requiring count forests. Every other panel is excluded by its acquired INITIAL constraints or a separately checked moment contradiction. The beta scopes are signed and are not identified by reflection.

Region	ρ interval	β interval	Method	Forests
B0	$[1, 11/10]$	full	resources/counts	24
BN	$[11/10, 23/20]$	full	resources/counts	32
B2 S0	$[23/20, 467/400]$	$[-b_2, 1/5]$	resources/counts	32
B2 S1	$[23/20, 467/400]$	$[1/5, b_2]$	resources/counts	18
B2 S2	$[467/400, 237/200]$	$[-b_2, 1/5]$	resources/counts	34
B2 S3	$[467/400, 237/200]$	$[1/5, b_2]$	resources/counts	20
B3 S0	$[237/200, 477/400]$	$[-b_3, 1/5]$	resources/counts	35
B3 T0	$[237/200, 477/400]$	$[1/5, 3/10]$	resources/counts	22
B3 T1	$[237/200, 477/400]$	$[3/10, b_3]$	resources/counts	9
B3 S2	$[477/400, 6/5]$	$[-b_3, 1/5]$	resources/counts	36
B3 T2	$[477/400, 6/5]$	$[1/5, 3/10]$	resources/counts	22
B3 lower	$[477/400, 6/5]$	$[3/10, 17/50]$	resources/counts	11
B3 corner	$[477/400, 6/5]$	$[17/50, b_3]$	gap-free	—

All intervals are closed. In particular the new boundary 17/50 belongs to both adjacent pieces. There are 295 gapped count forests in the complete cover. Each retains all six original nearest-count roots. All 1,770 roots are present, with 148,586 nodes and 75,178 leaves. The other 905 panel instances have acquired resource exclusions. The final verification record joins every row to its accepted proof objects.

For $\rho \in [6/5, 7361/4000]$, 41 closed Fourier strips give an upper bound below 4097/10000. Their six support regions are

ρ interval	Support radius R
$[6/5, 5/4]$	13/5
$[5/4, 7/5]$	117/40
$[7/5, 3/2]$	25/8
$[3/2, 8/5]$	133/40
$[8/5, 17/10]$	7/2
$[17/10, 7361/4000]$	15/4

The support has been acquired before these supported Fourier bounds are applied. The final finite-strip maximum is less than 0.409674. For the remaining tail, the published iid estimate gives

$$R_5(P) \le \frac{3031}{10000}\left(1+\frac{323}{500\rho}\right) \le \frac{819}{2000} = 0.4095 \quad (\rho \ge 7361/4000).$$

The exact endpoint substitution and monotonicity join the infinite tail without a missing boundary.

4 A direct sign-count inequality without a support gap

This section contains the new step. It applies to every standardized common iid marginal in its stated moment region, with no extremality, contact, bounded-support or predecessor premise.

Put $p = \mathbb{P}(X > 0)$, $q = 1 - p$, $m = \mathbb{E}X_+ = \mathbb{E}X_- > 0$, $v = pq$ and $h = v/m$. Zero atoms are assigned to the nonpositive class. Set

$$B = \mathbf{1}_{\{X>0\}}, \qquad Y = p + hX, \qquad W = Y - B.$$

Then $\mathbb{E}[W \mid B] = 0$. The two conditional variances are exactly

$$v_0 = p^2\left(\frac{q(1-\beta)}{2m^2} - 1\right), \tag{4}$$

$$v_1 = q^2\left(\frac{p(1+\beta)}{2m^2} - 1\right). \tag{5}$$

For $J = B_1 + \cdots + B_5$, conditional independence gives

$$\mathbb{P}(J = j) = b_j(p) := \binom{5}{j} p^j q^{5-j}, \qquad V_j := \mathrm{Var}\Big(\sum_i W_i \mid J = j\Big) = (5-j)v_0 + jv_1.$$

Conditioning on a particular sign vector gives independent conditional variables. Every vector with the same count j gives the same sum law, which justifies conditioning only on the count.

Lemma 3 (Gap-free count bound)**.** *Define*

$$T(V,d) = \begin{cases} 1, & d \le 0, \\ V/(V+d^2), & d > 0. \end{cases}$$

For every real z,

$$\mathbb{P}\Big(\sum_{i=1}^{5} Y_i \le z\Big) \le Q(z) := \min\left\{1, \sum_{j=0}^{5} b_j(p)\,T(V_j, j-z)\right\}. \tag{6}$$

Consequently, at $H = (z-5p)/h$, the normalized positive CDF error is at most

$$\frac{\sqrt{5}}{\rho}\left[Q(z) - \Phi\left(\frac{(z-5p)m}{\sqrt{5}\,pq}\right)\right]. \tag{7}$$

Proof. A centered random variable U with variance V obeys $\mathbb{P}(U \le -d) \le V/(V+d^2)$ for $d > 0$. For $V > 0$ this follows by applying Markov's inequality to $(U - V/d)^2$ on that event; for $V = 0$ the event is empty. At $d = 0$ the safe bound is one, including $V = 0$. Apply this separately to each conditional sum and use its actual binomial probability. Finally $\sum Y_i = 5p + hS_5$ gives the normal coordinate. □

The convention $T(0,0) = 1$ is essential. Substituting zero would omit a possible atom exactly on an integer count boundary.

4.1 Reducing the corner to two-dimensional boxes

The additional moment constraints are

$$m \ge \frac{1+\beta^2}{2\rho}, \qquad \frac{2m^2}{1+\beta} \le p \le 1 - \frac{2m^2}{1-\beta}, \qquad m^2 \le pq. \tag{8}$$

The first two follow from the signwise Cauchy inequalities already used. For the last, conditional second moments give $1 \ge m^2/p + m^2/q = m^2/(pq)$.

Consider a closed beta slab $[b_l, b_u]$ in $[17/50, b_3]$, keeping the whole moment interval $\rho \in [477/400, 6/5]$. Set

$$m_l = \frac{1+b_l^2}{2(6/5)}, \quad p_l = \frac{2m_l^2}{1+b_u}, \quad p_u = 1 - \frac{2m_l^2}{1-b_l}.$$

Every law lies in this p interval; all the selected intervals are contained in $(0, 1/2)$. For a closed subbox $p \in [l, u]$, $z \in [a, b]$, use

$$\bar{v}_0 = \max\left\{0, u^2\left(\frac{(1-l)(1-b_l)}{2m_l^2} - 1\right)\right\},$$
$$\bar{v}_1 = \max\left\{0, (1-l)^2\left(\frac{u(1+b_u)}{2m_l^2} - 1\right)\right\}.$$

These bound (4)–(5) above: each actual bracket is nonnegative, and each positive factor has been enlarged. A negative upper bracket means that the actual-law domain is empty; clipping it to zero remains safe. The exact upper binomial mass is attained at $p_j = \min(u, \max(l, j/5))$. Hence an upper CDF on this whole box is

$$\bar{Q} = \min\left\{1, \sum_{j=0}^{5} b_j(p_j) T\big((5-j)\bar{v}_0 + j\bar{v}_1, j-b\big)\right\}.$$

Let $\nu = a - 5u$. A lower bound for the normal argument is

$$x_l = \begin{cases} \nu m_l/(\sqrt{5}\, u(1-u)), & \nu \ge 0, \\ \nu/\sqrt{5u(1-u)}, & \nu < 0. \end{cases} \tag{9}$$

The first branch uses the lower mean and upper variance factor. In the second branch use $m \le \sqrt{pq}$; if the actual numerator is positive the negative lower bound is automatic. Otherwise note that for $z \ge 0$ and $p < 1/2$,

$$\frac{d}{dp}\frac{z-5p}{\sqrt{p(1-p)}} = -\frac{5p + z(1-2p)}{2[p(1-p)]^{3/2}} < 0.$$

Thus (9) holds on the entire box. When evaluating it exactly, the negative numerator is divided by a *lower* square root. Reversing that rounding direction would invalidate the bound.

The resulting whole-box score bound is

$$\frac{\sqrt{5}}{477/400} \max\{0, \bar{Q} - \Phi(x_l)\}.$$

Taking the positive part before multiplication avoids an incorrect upper-factor substitution on a negative difference.

4.2 The finite certificate and the infinite phase rays

The twelve closed beta slabs have endpoints

$$0.340, 0.341, \ldots, 0.351, b_3.$$

For each slab, the p interval is rounded outward to a 10^{-6} grid. The five phase intervals $[0,1], \ldots, [4,5]$ give 60 initial boxes. Recursive midpoint splits in p and z produce a complete forest with 1,994 nodes and 1,027 leaves. Exact acceptance recomputes every leaf and checks both children of every split. Its maximum stored score upper is

$$\frac{20484834187}{50000000000} = 0.40969668374 < 0.4097 < C_{\mathrm{E}}.$$

A separately implemented rational Gaussian evaluator checks every leaf; its maximum is 204848338979/500000000000, no larger than the stored bounds. It does not import the original numerical oracle.

For completeness let m_L, p_L, p_U denote the same moment bounds for the *whole* corner $\beta \in [17/50, b_3]$. Set

$$v_U = p_U(1-p_U), \quad s_U = v_U^2/m_L^2 - v_U, \quad V_j^U = \max\left\{\frac{j}{p_L}, \frac{5-j}{1-p_U}\right\} s_U.$$

Here $s = \mathbb{E}W^2 = v^2/m^2 - v \le s_U$: on $v \ge m_L^2$ the function $v^2/m_L^2 - v$ is increasing. Also $b_j(p) \le \binom{5}{j} p_U^j (1-p_L)^{5-j}$. For $z \le 0$, drop the nonnegative normal CDF and apply monotonicity at zero. This gives the uniform bound

$$\frac{\sqrt{5}}{477/400}\left[(1-p_L)^5 + \sum_{j=1}^{5}\binom{5}{j} p_U^j (1-p_L)^{5-j}\frac{V_j^U}{V_j^U + j^2}\right] < 0.331.$$

For $z \ge 5$, $F \le 1$ and the normal argument at five is $\sqrt{5}\, m/p$. Cantelli for the standard normal gives the bound

$$\frac{\sqrt{5}}{477/400}\frac{1}{1 + 5m_L^2/p_U^2} < 0.187.$$

Both displayed inequalities are verified over exact rationals with an outward square root of five. The two rays and the compact closed interval cover every real threshold.

Proposition 4 (Independently certified gap-free corner)**.** *For every standardized iid marginal with*

$$\rho \in [477/400, 6/5], \qquad \beta \in [17/50, b_3],$$

and every raw threshold H, the normalized positive CDF error is strictly less than 4097/10000. *No C_4, support, contact, or physical-gap assumption is needed for this proposition.*

5 What the other certificates prove

The remaining low-moment regions concern the attained contrary witness. Their chain is

$$\begin{aligned}\text{contact and } q_5 = 0 &\implies \text{support and Fourier bounds}\\ &\implies \text{INITIAL} \implies \text{resources} \implies \text{counts.}\end{aligned}$$

Each arrow has a continuous proof as well as a source-specific arithmetic obligation. INITIAL is a finite induction started with *no* physical defect or gap seed. All surviving final cells

must acquire a genuine central support gap; repeated improvement without that gate proves nothing about a gap.

The defect used in this induction is

$$D = (3+\sqrt{10})\rho - \gamma - 3J, \qquad \gamma = \mathbb{E}X^3, \quad J = \sum_{(a,b)} \tau_{a,b}(b-a)^2,$$

where the sum runs over bounded gaps of the support and $\tau(t) = \mathbb{E}[X\mathbf{1}_{\{X\geq t\}}]$ is constant on each gap. The independently derived conserved identity comes from integrating the same contact slack against τ in two ways and using compensated smooth Fourier inversion. It includes the complete high-frequency tail and keeps the half-open gap convention at atoms. Its proof does not use an asymptotic expansion in n.

Once a central gap $(-A, B)$ is acquired, the resource induction keeps bounds on the same marginal's moments, endpoint locations, and conditional noise. One small but necessary repair retains the already available price

$$6mAB \geq 6m_L ab > 0 \qquad (m \geq m_L,\ A \geq a,\ B \geq b).$$

It can be combined with another lower price by taking their maximum. Since $4AB \leq (A+B)^2$, a lower price P gives $A + B \geq \sqrt{2P/(3m_U)}$ when $m \leq m_U$. The new term alone satisfies

$$\frac{2(6m_L ab)}{3m_U} \leq 4ab \leq (a+b)^2.$$

It therefore retains a gap already proved; it does not manufacture a physical gap from a failed INITIAL calculation.

5.1 The actual iid count probability

For the same representation as in Section 4 and a nearest-count phase $z = k+u$, let W_j denote the sum of the five noises conditional on exactly j positive signs. The closed CDF obeys

$$\mathbb{P}(S_5 \leq H) \leq F_{\mathrm{Bin}(5,p)}(k) - b_k(p)\mathbb{P}(W_k > u) + \sum_{j>k} b_j(p)\mathbb{P}(W_j \leq u - (j-k)). \tag{10}$$

The outgoing event is *strict*, the incoming event is closed. Atoms on the event plane must not be subtracted twice. Incoming bounds use Cantelli or exponential estimates. An outgoing minorant follows from

$$G_a(x) = -x^4 + 3a^2x^2 + 2a^3x = x(x+a)^2(2a-x), \qquad a > 0.$$

It is nonpositive for $x \leq 0$ and has global maximum $(9/4 + 3\sqrt{3}/2)a^4$. Therefore

$$\mathbb{P}(U > u) \geq \frac{\mathbb{E}G_a(U-u)}{(9/4+3\sqrt{3}/2)a^4}.$$

Bounded support and common conditional moment constraints control the expectation. The two conditional third moments remain common variables through all constraints. Linear or positive-denominator affine-ratio objectives are maximized over the resulting bounded polygon at its vertices. This is a relaxation containing every actual law; feasibility of a relaxed point is never asserted to produce a convolution law.

The exact forest tests include the coupled moment and physical-defect constraints, phase polynomials and Gaussian enclosures. A leaf closes only through a proved emptiness predicate or an upper score strictly below the fixed rational lower enclosure of C_{E}. Every original nearest-count root $k = 0, \ldots, 5$ is present. Its domain is derived from acquired mean, scale and threshold bounds, so thresholds outside the original roots are not silently discarded.

6 Closing the contradiction

Proof of Theorem 1. Suppose $C_5 > C_{\mathrm{E}}$. Lemma 2 supplies a positive closed-edge maximizer, and (1) supplies its zero predecessor allowance. Its moments satisfy $\rho \geq 1$ and (3).

If $\rho \geq 6/5$, the complete Fourier cover and the cited tail bound exclude it. Otherwise it belongs to one of the closed rows in the low-moment table. Every gapped row is excluded by the fully acquired resource/count composition. The one remaining row is excluded directly by the gap-free proposition. All seams are covered, including $\rho = 477/400$, $\rho = 6/5$, and $\beta = 17/50$. Thus no contrary maximizer exists, and $C_5 \leq C_{\mathrm{E}}$. □

The proof needs only a single fixed sample size. It avoids the long local-domination and propagation chain used for eventual cutoffs. Combining this result with any separate theorem for other indices requires checking that theorem's own scope; it is not part of the present conclusion.

7 The finite continuation to six

Write G_n for the assertion $C_n < C_{\mathrm{E}}$, with the supremum in C_n taken over all centered, unit-variance laws having a finite third absolute moment. The finite continuation [5] establishes the implications below. Its analytic hypotheses are retained explicitly; they are not consequences of the finite arithmetic.

Assumption 5 (Retained finite-index inputs)**.** Let $n \in \{6, 7, 8, 9\}$.

(i) $\mathcal{A}_6$: the inherited LOW, physical-gap, resource, charged-score and count reductions hold with their stated same-law hypotheses. In particular, their finite acquisitions apply to an unrestricted maximizing witness on the contrary branch $C_n \geq C_{\mathrm{E}}$. The initial acquisition assumes no defect seed: each stage covers its whole closed moment domain and uses only previously established bounds. The resulting probability panels and all $n+1$ original count roots cover every surviving witness, and the acquired count majorant bounds that witness's normalized score.

(ii) $\mathcal{F}$: the smoothing inequalities, profile bounds and elementary enclosures used in the supported Fourier estimates hold uniformly on their stated domains, with their stated remainder bounds. This includes the literature inputs, not merely evaluations at selected parameter values.

The separate tail hypothesis $\mathcal{T}_{10}$ is

$$C_n < C_{\mathrm{E}} \qquad \text{for every integer } n \geq 10.$$

Here $\mathcal{T}_{10}$ denotes the earlier conditional continuation, including its retained analytic and historical certificate premises.

Fixed-index attainment, reflection at an atom, the contact inequality and the continuous support criterion have separate analytic proofs. They are distinguished from the retained LOW and count reductions in Assumption 5. The coarse global estimate

$$C_5 < \frac{429}{1000} \tag{11}$$

is acquired independently of G_5, within the same analytic and Fourier framework. No sharp fifth-index conclusion is used to initiate the continuation.

Theorem 6 (Conditional finite continuation)**.** *Under* $\mathcal{A}_6$ *and* $\mathcal{F}$*,*

$$G_6 \wedge G_7 \wedge G_8 \wedge G_9.$$

Under the additional hypothesis $\mathcal{T}_{10}$*,*

$$C_n < C_{\mathrm{E}} \qquad (n \geq 6).$$

Proof. Suppose first that $C_6 \geq C_{\mathrm{E}}$. Attainment and whole-law reflection give a maximizing law and a positive CDF branch at a finite raw threshold. Reflection interchanges the right CDF with the corresponding left limit; atoms require no continuity assumption. All subsequent acquisitions refer to this one unrestricted maximizing law.

The contact allowance is licensed by (11):

$$6\big(R_5(P) - C_6\big)_+ < 6\left(\frac{429}{1000} - \frac{4097}{10000}\right) = \frac{579}{5000}, \qquad \frac{4097}{10000} < C_{\mathrm{E}}.$$

Use $q_6 = 579/5000$ in the support criterion. Its full-ray estimates are uniform in the raw threshold. The nine acquired closed moment ranges cover $[1, 1.84025]$; support is established before each supported Fourier estimate is applied. The supported Fourier bounds exclude $1.185 \leq \rho \leq 1.84025$ by the ceiling $4097/10000 < C_{\mathrm{E}}$. Shevtsova's inequality [3] gives, for $\rho \geq 1.84025$,

$$R_n(P) \leq 0.3031\left(1 + \frac{0.646}{\rho}\right) \leq 0.4095 < C_{\mathrm{E}}.$$

Thus the remaining closed bands, with support acquired before its use, are

Band	Third moment	Support
B_0	$[1, 11/10]$	$\lvert X\rvert \leq 23/10$
B_N	$[11/10, 23/20]$	$\lvert X\rvert \leq 12/5$
B_2	$[23/20, 237/200]$	$\lvert X\rvert \leq 5/2$

Every moment seam belongs to both adjacent closed domains.

Within these bands the complete INITIAL acquisitions yield the physical gap and resource bounds. Each of the 100 closed probability panels is then excluded either by a whole-panel contradiction or by complete count forests. Every original root is retained; subdivisions cover both closed children. Each terminal score bound is strictly below a fixed rational lower bound for C_{E}. Assumption 5 therefore excludes the maximizing witness. Hence G_6.

Proceed successively at $n = 7, 8, 9$. On the contrary branch $C_n \geq C_{\mathrm{E}}$, the already proved global predecessor G_{n-1} gives, for the same maximizing law,

$$n\big(R_{n-1}(P) - C_n\big)_+ = 0.$$

Use $q_n = 0$ in that index's own support acquisition, Fourier bound, INITIAL acquisition and count cover. The preceding argument applies separately at each index. It includes equality $C_n = C_{\mathrm{E}}$, uses no monotonicity in n, and proves G_7, G_8, G_9, in that order. Attaching $\mathcal{T}_{10}$ proves the second assertion. □

An elementary compression of the finite cover

Lemma 7 (Sign probability)**.** *If* $\mathbb{E}X = 0$*,* $\mathbb{E}X^2 = 1$*, and* $\rho = \mathbb{E}|X|^3 < \infty$*, then*

$$\mathbb{P}(X > 0)\rho^2,\ \mathbb{P}(X < 0)\rho^2 \geq \kappa := \frac{8}{27}(7\sqrt{7} - 17) > \frac{11261}{25000}.$$

Proof. Put $p = \mathbb{P}(X > 0)$, $m = \mathbb{E}X_+ = \mathbb{E}X_- > 0$, and $a = \mathbb{E}X_+^2 \in (0,1)$. Cauchy on each sign gives

$$m^2 \le pa, \qquad a^2 + (1-a)^2 \le m\rho,$$

and consequently

$$p\rho^2 \ge \frac{(2a^2 - 2a + 1)^2}{a}.$$

The derivative of the expression on the right is $(2a^2 - 2a + 1)(6a^2 - 2a - 1)/a^2$; its minimum is attained at $a = (1 + \sqrt{7})/6$ and equals κ. Reflection gives the other sign. An atom at zero does not affect either inequality. □

For $\rho \le r$, this confines $p = \mathbb{P}(X > 0)$ to

$$\frac{11261}{25000r^2} \le p \le 1 - \frac{11261}{25000r^2}.$$

Strict endpoint comparisons discard whole closed probability panels. The residual resource exclusions have two terminal forms. In the first, acquired nonnegative coefficients and a physical defect give

$$D \ge d \ge a(1-\theta)^{3/2} + b\theta, \qquad 0 \le \theta \le 1, \qquad a > D, \quad b \ge \tfrac{3}{2}a.$$

Convexity makes the right side at least $a + (b - 3a/2)\theta \ge a > D$, a contradiction. In the second, the acquired rational bounds $\rho_- \le \rho \le \rho_+$ satisfy $\rho_- > \rho_+$. The complete four-index cover is

	Sign	Tangent	Inconsistent ρ	Count forests	Total
$n = 6, 7, 8, 9$	824	52	40	284	1200

The count forests have 2,362 original roots and 257,089 leaves. This compression removes redundant exposition, not the acquisitions used inside the surviving resource and count arguments.

Dependency boundary

The sharp fifth-index argument and the finite continuation have different roles. Their logical arrangement is

$$\begin{gathered}
\text{published inequalities} + \mathcal{A}_6, \mathcal{F} \Longrightarrow C_5 < 429/1000 \\
\Downarrow \\
G_6 \Longrightarrow G_7 \Longrightarrow G_8 \Longrightarrow G_9 \\
(G_6 \wedge \cdots \wedge G_9) + \mathcal{T}_{10} \Longrightarrow (G_n)_{n \ge 6}, \\
\text{separately: fifth-index argument, with its own inputs} \Longrightarrow (C_5 \le C_{\mathrm{E}}).
\end{gathered}$$

The right-hand chain uses only the coarse fifth-index seed, not the sharp fifth-index bound. Conversely, the fifth-index argument does not require this tail branch. An eventual or finite-tail theorem cannot replace a missing fifth-index proof. Theorem 1 proves $C_5 \le C_{\mathrm{E}}$, not the strict assertion G_5.

8 Continuous reductions and arithmetic certificates

The proof of Theorem 1 consists of the following implications:

$$C_4 \le C_{\mathrm{E}}, \quad C_5 > C_{\mathrm{E}} \implies \text{attained positive closed witness},$$
$$\text{contact with } q_5 = 0 \implies \text{support before higher moments},$$
$$\text{support and the same-law Fourier bounds} \implies \text{finite moment and probability covers},$$
$$\text{gapped count bounds or the direct gap-free comparison} \implies \text{contradiction}.$$

The direct gap-free proposition is valid for arbitrary laws in its moment rectangle and has no incoming contact or support edge. The conditional tail in Section 7 has no incoming edge into this fifth-index proof.

The companion proof supplement [4] supplies the continuous support criterion, compensated Fourier inversion and conserved identity, the complete LOW and HIGH inequalities, the finite resource induction and the conditional count majorants used in the gapped rows. These derivations are part of the proof, not consequences of a successful numerical calculation. Every quantitative acquisition is tied to the same unrestricted contrary witness and to its original closed parameter domains.

For each accepted count forest, exact acceptance checks every initial root, both children of every split and every terminal inequality. A terminal region is either necessarily empty or has a rational score upper bound below a fixed rational lower enclosure of C_{E}. Thus the finite trees prove inequalities over continuous domains, not over sampled points. Square roots are enclosed by integer squaring. Gaussian, exponential and trigonometric values use finite series with explicit remainder bounds. The gap-free corner has, in addition, a separate rational evaluation of every leaf and both unbounded rays.

The mathematical derivations, source code and complete finite certificates are available in the accompanying repository. The two archived source packages [4, 5] retain their distinct theorem scopes. In particular the conditional tail hypotheses of Assumption 5 are not discharged merely by adjoining the fifth-index certificate.

AI assistance declaration

ChatGPT and Codex were used in developing the mathematical arguments and proofs, constructing and checking computational certificates, and drafting and revising the manuscript. This included the GPT-5.6 Sol, GPT-6.1 Sol and GPT-6 Astra models. Kacper Rodziewicz and Bartosz Kołodziejek remain responsible for the mathematical claims and final text.

References

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[4] K. Rodziewicz and B. Kołodziejek, *The fifth iid Berry–Esseen constant: continuous reductions and exact finite certificates*, proof supplement, 20 October 2026. https://github.com/rodziewiczk/berry-esseen-045-certificate/releases/tag/cutoff5-sources-2026-10-20.

[5] K. Rodziewicz and B. Kołodziejek, *Conditional finite continuation at indices six through nine*, proof supplement, 20 October 2026. The retained analytic and tail hypotheses are stated in Section 7. https://github.com/rodziewiczk/berry-esseen-045-certificate/releases/tag/cutoff5-sources-2026-10-20.